PROCESS OR MATERIALS REVIEW REQUEST FORM

0Title: Additional organosilanes for biosensing research **Request number: #SEN_001**

Requestor: Ateeq Suria **Date: 05/16/2013**

Non-SNF staff, please complete the following:

Badger ID: asuria@stanford.edu **Email: asuria@stanford.edu** **Phone: 520-668-9830**

Company name/PI: Prof. Debbie G. Senesky

This request includes the material review request for 5 different chemicals.

1. **The chemical or material.** Please provide all common names, trade names, and CAS numbers where appropriate. Include an MSDS, if available; or provide the reason, if not. Make sure to include information for any new secondary chemicals (such as a developer for a new resist). Read the MSDSs as well as the Stanford Chemical Storage Groups and the Stanford Chemical Safety Data Base sections on this website to determine the Storage Group Identifier and Main Hazard Class of your chemical/material.

    - **Chemical 1:** Trimethoxy(octadecyl)silane.
        - Other names: ODTMS, Octadecyltrimethoxysilane
        - MSDS (hyperlinked) – attached to end of request
        - CAS Number: 3069-42-9
        - Storage Group: L
        - Main Hazard Class: Irritant
    - **Chemical 2:** Trichloro(octadecyl)silane
        - Other names: OTS, Octadecyltrichlorosilane
        - MSDS (hyperlinked) – attached to end of request
        - CAS Number: 112-04-9
        - Storage Group: B
        - Main Hazard Class: Flammable Liquid

- **Chemical 3:** Butylamine (secondary chemical for attachment of ODTMS). Works as a catalyst in deposition of the organosilane layers.
    - Other names: 1-Aminobutane, n-Butylamine
    - MSDS (hyperlinked) – attached to end of request
    - CAS Number: 109-73-9
    - Storage Group: A
    - Main Hazard Class: Flammable liquid
- **Chemical 4:** Toulene (secondary chemical for attachment of ODTMS, ODMS, OTS). Solution to attach organosilane is made of 0.1 to 2.5% organosilane, balance toluene.
    - Other names: None
    - MSDS (hyperlinked) – attached to end of request
    - CAS Number: 108-88-3
    - Storage Group: L
    - Main Hazard Class: Flammable liquid
- **Chemical 5:** Glutaraldehyde (cross-linker molecule). Comes as 25% aqueous solution
    - Other names: None
    - MSDS (hyperlinked) – attached to end of request
    - CAS Number: 111-30-8
    - Storage Group: G
    - Main Hazard Class: Toxic Solid or Liquid

2. **Vendor/manufacturer info:** address and phone number, website URL.

   Vendor for chemicals is Sigma Aldrich. 3050 Spruce Street, Saint Louis MO 63103 for Chemicals 1,2,3 & 5. Chemical 4 will be purchased from Stanford BioStores.

   Phone number: 1-800-325-5832

   ODTMS (Chemical 1):

https://www.sigmaaldrich.com/catalog/product/aldrich/376213?lang=en&region=US

OTS (Chemical 2):
https://www.sigmaaldrich.com/catalog/product/aldrich/104817?lang=en&region=US

Butlyamine (Chemical 3):
https://www.sigmaaldrich.com/catalog/product/sial/471305?lang=en&region=US

Toulene (Chemical 4):
https://www.sigmaaldrich.com/catalog/product/sial/179965?lang=en&region=US

Glutaraldehyde (Chemical 5):
http://www.sigmaaldrich.com/catalog/product/sial/g6257?lang=en&region=US

3. **Reason for request:** Please give serious thought to this. If you have any process information (application notes from the vendor, protocol from another lab, experimental methods section of an article), please include it, preferably as attachments. Ask yourself these questions: Is this the latest procedure? Are there newer/safer alternatives that will also work for my project? Will any of the current SNF approved chemicals and materials work for me?

   There are several reasons for this chemicals request. All of these have do to with providing an interfacial layer for attachment of biological molecules on semiconductor surfaces. Organosilanes provide this interfacial layer. They deposit as monolayers on top of hydroxylated surfaces to provide sites to covalently bond biological molecules to the surface substrates. This is a key step in the development of biosensors. Currently I have been working with APTES as the organosilane layer. APTES does not provide the right complementary groups to allow covalent bonding of the type of bacteria I am working with: *Deinococcus radiodurans.* It has been shown in literature that glutaraldehyde provides additional cross linker groups for attachment of proteins and biomolecules. Thus one of the additional chemicals is glutaraldehyde (Chemical 5).

   To investigate the impact of using different types of organosilane layers and their characteristics of attachment, I am requesting these additional organosilane compounds to be allowed into the SNF. The process flow I will use is described in Section 4. Another key step in the realization of this biosensor is to selectively deposit organosilane layers onto the substrate, so that it can be limited to the area of the transistor gate. To define this area standard photolithography steps are used.

   As part of the project, an intern of SNF REU program arrived on June 2nd. The intern's project focuses on patterning different types of organosilane layers on GaN. The ability of

attaching the bacteria will be assessed on the different organosilane layers. This attachment step will be performed outside the SNF in our bio facility. The intern's responsibility will be to perform the vapor deposition of these chemicals on GaN wafers. I will be the primary person working with the liquid deposition of these organosilanes.

The procedures I am using are the most up to date procedures. I have attached some papers which show how this process is done along with the chemicals that are needed. The papers are referenced at the end of this section. Both the papers in the experimental section show the process of first activating the OH groups on the surface of the GaN, and then deposition in a solution of the organosilane (OTS/ODTMS) with toluene. For ODTMS 2% butylamine is used as a catalyst to speed up the process. Since the process depends on these specific organosilanes and the compounds that are used to deposit these monolayers, there are no substitutes for them available in the SNF. Glutaraldehyde is attached to an APTES immobilized layer by deposition in a solution with water too.

Below is also Table 1 and 2 which describes the deposition procedure for different types of organosilanes in the form of a solution and using a vapor phase deposition device, respectively.

TABLE I. Several different protocol parameters for liquid phase organosilane deposition.

Organosilane	Substrate	Surface Preparation	Solvent	Concentration	Time	Notes	Reference
APTES/APTMS	Silica gel	NS	Toluene	10%	30 min		52
	Si	Piranha	Ethanol	100 mM	24 h		53
	Si	Piranha	Unspecified	2%	20 min	20 min bake at 120 °C	3
	Glass	Unspecified	IPA	5%	Unspecified	5% water added, at 60 °C	54
	Glass	Methanol, HCL, H_2SO_4, NaOH	Water	0.40%	2 h	PH adjusted to 3.0 at 75 °C	35
	PDMS/glass	Oxygen plasma	Acetone (95%)	2%	2 h		55
	Si/SiO_2	E-Beam lithography, HF	Ethanol (95%)	2%	1 h	Second step in patterning	5
	ZnO/SiO_2	Sonication in $NH_4OH:H_2O_2:H_2O$ (1:1:10)	Ethanol (95%)	4%	4-5 h		36
	Si	$NH_3:H_2O_2:H_2O$ (1:1:5) and $H_2SO_4:H_2O_2:H_2O$ (1:1:5)	Toluene	10%	1 h		56
	PDMS/glass	$HCL:H_2O_2:H_2O$ (1:1:5)	Ethanol	50%	30 min	In a microfluidic channel	57
MPTS	SiO_2	Nitric acid, Piranha	Benzene	5 mM - 40 mM	40 min	In N_2 atmosphere	58
	ZnO	Ion beam milling	Toluene	4%	24 h		25
	Si/SiO_2	Piranha, $HCL:H_2O_2:H_2O$ (1:1:6)	Benzene	1-2 mM	3 h		16
OTMS	SiO_2	Chromic/sulphuric acid	Toluene	5%	1.5 h	n-Butylamine catalyst	4
	Si	Piranha	Toluene	50 mM	4 h		59
	$LiNbO_3$, Quartz	Oxygen plasma	Hexanes	1% (24 mM)	4 h		34
	Si	Piranha	Bicyclohexyl	100 mM	24 h		60
OTS	Mica	HCL clean, dehydration, controlled rehydration	Cyclohexane	3 mM	20 min	Subsequent 2 h bake at 120 °C	28
	Si	Piranha	Bicyclohexyl	100 mM	24 h		60
	Si	Piranha, HF, UV ozone	Hexane, hexadecane or toluene	1-50 mM	30 s	Compared with microcontact printing	42
	Si	Piranha	Toluene	0.05-1 mM	Varied		61
	Si	Sonocation, HF, Piranha	Bicyclohexyl	1-10 mM	24 h	Water added to solvent for some depositions	62
	SiO_2	Oxygen plasma	Toluene	0.50%	16 h	Baked at 120 °C for 30 min	7
	$LiNbO_3$	IPA sonication	Toluene	100 μM	Overnight	Rinsed with toluene and baked (1 h, 110 °C)	41
	$LiNbO_3$, Quartz	Oxygen plasma	Hexane: Chloroform (4:1)	0.1% (2.4 mM)	4 h		34
	$LiNbO_3$	Oxygen plasma	n-Hexane	1 mM	1 h		15
	Si/SiO_2	Piranha	Unspecified	Unspecified	Unspecified		43
PEO	TaO_5	UV/Ozone	Toluene	3 mM	18 h	0.08% HCl	8
PEG	PDMS	Oxygen plasma	Toluene	0.1 M	30 min		49
PFS	Si	Piranha	Iso-octane	5 mM	10 min		44
	$LiNbO_3$	IPA sonication	Toluene	100 μM	Overnight	Rinsed with toluene and baked (1 h, 110 °C)	41

APTMS—(3-Aminopropyl)trimethoxysilane, APTES—(3-aminopropyl)triethoxysilane, MPTS—(3-mercaptopropyl)trimethoxysilane, OTS—octadecyltrichlorosilane, OTMS—Octadecyltrimethylsilane, PFS—polytetrafluoroethylene based organosilanes, PEG—poly(ethylene glycol) based organosilane, PEO—poly(ethylene oxide) based organosilane.

036501-4 Glass *et al.* Biomicrofluidics **5**, 036501 (2011)

TABLE II. Several different protocol parameters for vapor phase organosilane deposition.

Organosilane	Substrate	Surface preparation	Temperature	Pressure	Time	Notes	Reference
APTMS	Glass	Piranha	Unspecified	670 Pa	16 h		11
MPTS	SiO_2	Sulphuric acid/sodium peroxydisulphate	100 °C	Unspecified	Unspecified	Also pumped (pressure unspecified)	63
	Si	Piranha, oxygen Plasma	Unspecified	1 mTorr	4 h		64
	Silica	Piranha, UV ozone	Unspecified	200 mBar	0-120 h		39
OTMS	SiO_2	UV irradiation	100/150 °C	Unspecified	1-16 h	Patterned with AFM	27
OTS	Si	UV irradiation	150 °C	Unspecified	0.5-8 h		26
PFS	Si	UV irradiation	150 °C	Unspecified	0.5-8 h		26
	Si	$NH_4O_4/H_2O_2/H_2O$ (1:1:4)	45 °C	Unspecified	3 h	In N_2 atmosphere	5
	SiO_x	Plasma	Unspecified	1 Torr	1 h		48
	SU8	Unspecified (after patterning)	Unspecified	0.5 bar	Unspecified		46

APTMS—(3-Aminopropyl)trimethoxysilane, MPTS—(3-mercaptopropyl)trimethoxysilane, OTS—octadecyltrichlorosilane, OTMS—octadecyltrimethylsilane, PFS—polytetrafluoroethylene based organosilanes.

To bring up the discussion on alternatives, instead of depositing using a solution of organosilane with toluene, we can use molecular vapor phase deposition to deposit the monolayers. The Savannah tool provides this option. The use in the savannah is dependent upon the chemical composition and boiling points of this liquid. Sigma Aldrich and Gelest have provided quotes confirming that the organosilanes can be filled into a Swagelok 50ml cylinder enabling use in the Savannah. This part of the request is pending the results of the CV characterization experiments which will provide information on contamination levels pre and post deposition of APTES and eventually other organosilanes such as OTS and ODTMS. If the results show that organosilanes are being deposited then the additional precusors will be ordered when the MVD tool currently being installed.

References for Protocols

Baur, B. et al. Chemical functionalization of GaN and AlN surfaces. Applied Physics Letters 87, 263901 (2005).

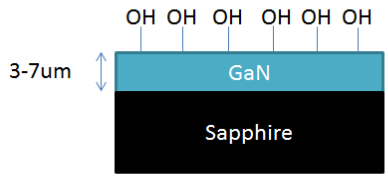
Schoell, S. J. et al. Functionalization of 6H-SiC surfaces with organosilanes. Applied Physics Letters 92, 153301 (2008).

Glass, N. R., Tjeung, R., Chan, P., Yeo, L. Y. & Friend, J. R. Organosilane deposition for microfluidic applications. Biomicrofluidics 5, 36501–365017 (2011).

4. **Process Flow:** Please provide a detailed process flow description on how and where you proposed to use this chemical. This should include **all** **Lab equipment** to be used for processing your wafers once your new chemical or material has been used (even if your new material is a film that is removed, it may still pose potential contamination concerns.) Make sure to include wet benches. Please note that if the chemical/material is to be used in any the "clean" equipment, purity specifications will be needed. This is most important for chemicals/material that are not normally used for VLSI device fabrication. To be allowed into a "clean" tool, the material should MOS grade or better.

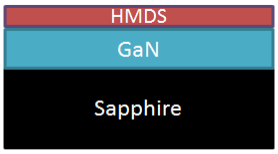
The process flow that will be used is described below step by step for processing of the tool.

Step 1: Cleaning & Oxidation of GaN



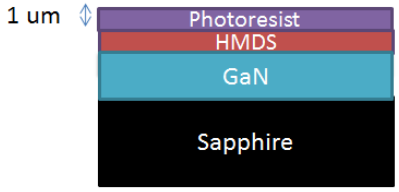
Done at wbgaas.
Clean in Pirhana solution at wetbench for 1 minute. Wash with DI water and dry with N2.
Thickness of GaN – 3 to 7 µm

Step 2: HMDS Deposition



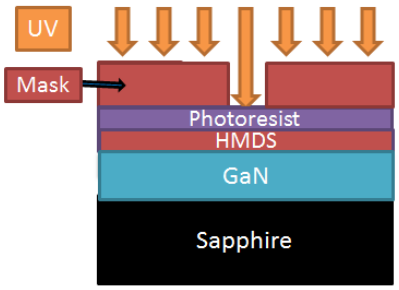
Dehydrate on 90C oven for 5 min.
Place in YES oven to deposit HMDS.

Step 3: Photoresist Deposition



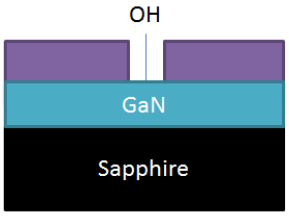
Spin Shipley 3612 using Headway. 60 seconds at 5000 rpm. Then prebake on hot plate 90C for 60sec.
Thickness of PR – 1 µm

Step 4: Mask Alignment



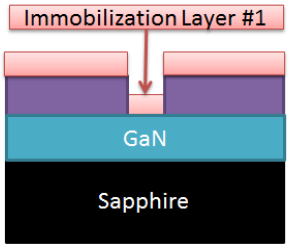
Align mask using karlsuss
Expose for 1 sec, 15 mW/cm2
Post bake 115C for 60sec.

Step 5: Development



Develop in MF 26-A for 60s
Rinse with DI water, dry with N2
Hard bake – 2 step
1. 115C for 15min
2. 200C for 15 min

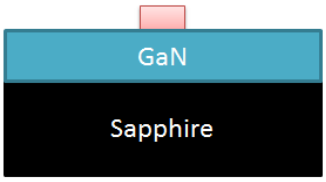
Step 6: Organosilane #1 Deposition



On savannah run recipe to clear the exhaust gases and equilibrate
Next run organosilane #1 (ODMS, ODTMS, OTS) recipe to deposit monolayer. Note $T_{savannah} < T_{hard_bake}$

This is the first step where the new chemicals being requested will be used.

Step 7: Liftoff



Perform liftoff by placing in acetone for 5 minutes. Wash with DI water and dry with N2 gas.
Done at wbsolvent.
If sonicator needed, at wbgen.

The equipment group we will be using is gold-contaminated. To avoid contamination at solvent benches and wet benches we will be using our own beakers. The procedure has been updated to only include deposition of one type of organosilane layer at a time.

After the completion of step 5 and the hard bake at 115°C, there are two options to proceed as stated above: liquid and vapor phase deposition. Both these protocols are described in detail below depending on the type of chemical being used. All practices have been discussed with Mary Tang and Michelle Rincon in detail to ensure safe practices and safety of other lab members.

There are several different safety measures that are being taken to ensure safe practices. Most of the solutions that will be prepared and used will be in very dilute concentrations. For disposal standard procedures will be followed and if standard procedures do not apply (such as in the case of glutaraldehyde) then a hazardous waste pickup will be requested.

Additionally, standard PPE will be used such as yellow chemical resistant Nitril gloves, chemical aprons and face shields when handling or preparing the chemicals and their solutions. Additionally if needed a full face respirator will be used.

Glutaraldehyde Attachment to APTES layer

Bench: wbgaas

Using the standard recipe developed with consultation from Michelle I will be depositing APTES using the Savannah tool as specified above and described in the process flow up to Step 6. Before performing Step 7 an additional step will be performed. This cross-linker molecule provides an additional surface for biomolecules to bind to. Since glutaraldehyde is known to bond well to proteins typically found in the cell wall of bacteria, I want to use this additional cross-linker layer to attempt to attach my bacterial molecule. To perform this deposition a 2.5% v/v solution of glutaraldehyde solution will be used. This solution will be prepared from an already dilute aqueous solution of glutaraldehyde. One protocol to be used will involve mixing of 20 ml of DI water with 74 µl of glutaraldehyde. The wafers after processing in step 6 will be placed for 1 hour at room temperature in this solution after which it will be washed with DI water and blown dry with nitrogen gas. This procedure will be done at the wbgaas. The disposal of glutaraldehyde has been described in detail below in Section 7. The glutaraldehyde ordered will arrive in a 25% aqueous solution form and will be diluted down by adding to water as described above to make is a 2.5% v/v solution. Additionally I have shadowed a fellow PhD student who performs a similar protocol in his lab which is a non-clean environment.

Liquid Deposition Protocols

ODTMS

Bench: wbsolvent

As part of Step 6, the photoresist patterned GaN chip will be left in a solution of 5% ODTMS (6ml) in toluene (120ml) with 0.5% of butylamine (0.6ml) added as a catalyst. This deposition takes 60 minutes at room temperature conditions. The procedure in its entirety will be carried out at the wbsolvent in the SNF. Once this step has been completed, then liftoff will be performed as detailed in Step 7 at the wbsolvent also. This ODTMS will be purchased as a concentrated solution as available from Sigma and then diluted down at one of the wet benches. It comes in a 90% assay from Sigma.

Butylamine

Bench: wbsolvent

Butylamine will be diluted down solution in toluene from a concentrated solution. It will be diluted down and stored in the storage bottles described above. At time of use the solution will be diluted down further to the concentration used in the process above as

a catalyst accordingly. It comes as a 99.5% assay from Sigma.

OTS

Bench: wbsolvent

As part of Step 6, the photoresist patterned GaN chip will be left in a solution of %0.1 (v/v) OTS in toluene. This deposition takes 30 minutes at room temperature conditions. The procedure in its entirety will be carried out at the wbsolvent in the SNF. Once this step has been completed, then liftoff will be performed as detailed in Step 7 at the wbsolvent also. Similar to the ODTMS, the OTS will be prepared by diluting down the solution obtained from Sigma. Similar to ODTMS this is comes in a 90% assay from Sigma.

Vapor Deposition Protocols

These protocols will be very similar to the one currently used for APTES. The general procedure used is as follows. The changes would be the precursor that is used in Step VII.

Standard Protocol for Organosilane Deposition in *Savannah*

1. Run HfO_2-Standard for 25 cycles to coat the chamber ~20 minutes
2. Run Organosilane Deposition Recipe ~ 1.5 to 2 hrs
    I. Set pressure threshold at 300 Torr
    II. Flow at 20 sccm
    III. Heater 8 at 100° C (inner chamber)
    IV. Heater 9 at 100° C (outer chamber)
    V. Heaters 6 at 150° C (stop valve)
    VI. Heaters 7 at 150° C
    VII. Heater Organosilane Precursor at 60° C
    VIII. Stabilize 8 (inner chamber)
    IX. Stabilize 9 (outer chamber)
    X. Flow 10 sccm
    XI. Stop valve 0 (close stop valve)
    XII. Pulse Organosilane precursor for 5 seconds
    XIII. Wait 20 seconds
    XIV. Stop valve 1 (open stop valve)
    XV. Repeat from Step 9 for (input #) number of cycles
    XVI. Flow 20 sccm
    XVII. Wait 60 seconds
    XVIII. Flow 5 sccm
    XIX. Reset Pressure Threshold back to 250 Torr
3. Run HfO_2-Standard for 100 cycles to clean the chamber at 200°C ~45 minutes

ODTMS

The deposition for ODTMS will happen as described above with Step 2.VII and Step 2.XII. The remainder of the procedure will be the same including all the temperatures.

OTS

The deposition for OTS will happen as described above with Step 2.VII and Step 2.XII. The remainder of the procedure will be the same including all the temperatures.

5. **Amount and form.** How much will you bring in? Is it solid, powder or liquid? (Note: as a general rule, powders are not permitted in the cleanroom.) Do you need to mix it to use it?

   The chemical will be brought in two forms: liquid and as ALD precursor canisters.

   Chemical 1 & 2: Liquid Form. Bring only 25ml at a time. The canisters will come in the Swagelok 50mL cylinders that are used for the Savannah. Sigma can provide these cylinders.

   Chemical 3: Liquid form. 25 ml. Comes in bottles of 25ml, 250ml, 1L, etc.

   Chemical 4: Liquid form. 500ml. Comes in many different sizes.

   Chemical 5: Liquid form. 100ml. It comes as a 25% aqueous solution.

6. **Storage:** Will you be storing your chemical/material at SNF? If so, please note any potential reactivity's (this should be on the MSDS). Storage groups A, B, D and L are stored in the yellow solvent cabinet in the furnace support area, while storage groups C, E, F and G are stored on top of one of the Pass-through Carts. Ensure your chemical container or material is properly labeled. If there is no available room, it must be stored by in the bulk storage area. You will then need to obtain it from receiving area personnel each time you want to use it and return it to them when you are finished using it (or each time you leave the lab). Note that there is no storage of chemicals/materials in the processing lab or at any wet bench.

   The storage groups that are specified for Chemicals 1,2,3 and 4 is either A, B or L which all come under the yellow solvent cabinet storage cabinet. Thus these chemicals will be stored at the SNF in that solvent cabinet in the furnace support area. Chemical 5 comes under storage group G so it will be stores on top of one of the Pass-through Carts. It was further discussed that since these chemicals are moisture sensitive and since that might pose an issue, we will store the chemicals temporarily while in use at the nSiL in a nitrogen dessicator that either the SNF will provide or one will be purchased by us for use.

   As for the ALD precursor canisters, the storage will be done in the yellow cabinet for ALD canisters as per the regulations and standards for other ALD precursor storage.

Before bringing the chemicals into the lab, the containers will be tagged by Mahnaz.

7. **Disposal**: How will you dispose of any waste or excess chemical or material? In your discussions with experts and vendors, try to determine the best way to dispose of your spent chemicals and by-products. Please refer to the SNF Lab members Safety Manual for the different methods of waste disposal that are available in the lab.

Excess waste will be disposed based on the type of chemical being used. As advised by the committee all the additional chemicals will be disposed in a similar manner to that of glutaraldehyde below. All containers will be clearly marked for hazardous waste diposal.

For the Glutaraldehyde chemical, it will be disposed of as advised by the EH&S. It is to be placed in a container and marked as hazardous waste. This will then be placed in the hazardous waste area and EH&S will be contacted/informed to arrange a pickup for proper disposal.

To be completed by the PROM COMMITTEE:

Special handling requirements:

On-going approval or staff interaction requirements:

Approved By/Date:

This form is for requests for new materials, unusual processing. Submit for to: snf-promcommittee@stanford.edu. Please be as detailed as possible in filling out the form, answering all questions thoroughly. Additional supporting materials are welcome and in some cases necessary.